

What phase-locked template subtraction does to the heartbeat-evoked potential, and what a QRS-limited template costs

Companion note to the CFA ear-EEG paper

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Summary. A cardiac-artifact template indexed by heartbeat phase cannot distinguish the artifact from a cortical response that is itself phase-locked to the heartbeat. The heartbeat-evoked potential (HEP) is such a response, and it is reported at roughly 200–600 ms after the R-peak—inside the template. This is not a risk to be hedged; it is a certainty, and the only useful question is how much is at stake. This note measures it on the same behind-the-ear recordings used in the paper, and quantifies the alternative: restricting the template’s support to the QRS interval, which leaves the HEP interval untouched at a measured cost in artifact suppression.

1 The identifiability statement

Write the measurement as $y = s_a + s_p + d_r + n$, splitting the neural signal into a cardiac-asynchronous part s_a and a part s_p that is phase-locked to the cardiac cycle. The paper’s estimator observes only y and the R-peak instants $\{t_k\}$, and returns a template $\hat{m}(\phi)$ that is a function of cardiac phase alone. Both d_r and s_p are functions of ϕ alone. They are therefore *not identifiable* from $(y, \{t_k\})$: no learning rule, gain schedule, or amount of data separates them, because they are the same object as far as the observation model is concerned.

Two consequences follow, and they should be stated together rather than traded against each other. Any phase-locked cortical activity is removed along with the artifact. And a method that avoided removing it would, by the same argument, have to leave the phase-locked part of the artifact behind. Separating the two requires information the timing-only estimator does not have—a second channel with a different mixing, a stimulus-locked contrast, or a model of one of the two components.

2 Measurement

We use the two-channel behind-the-ear EEG of SeizeIT2 (OpenNeuro [ds005873](#)), high-pass filtered at 0.5 Hz, notch filtered at 50 Hz, resampled onto the ECG grid, with only supervisor-approved learnable beats entering the average. The analysis window $[-50, 600]$ ms relative to the R-peak is split into a *QRS part* $[-50, 200)$ ms and an *HEP part* $[200, 600)$ ms, which partition it exactly, so the energy share attributed to each is a genuine share.

Since the true disturbance is unobservable on real recordings, the phase-locked content of each interval is measured against a null that shares the record’s statistics but destroys phase locking. Let $\mathcal{E}_P[z]$ be the energy of the P -triggered average of z over the interval of interest. Averaging on the true peaks gives $\hat{m} = m + \ell$ with leakage ℓ of scale $\text{SD} / \sqrt{N_r}$; a circular time-shift of the whole peak train preserves the RR-interval sequence but destroys absolute phase, so it yields leakage only.

Averaging 200 such shifts gives the null floor $\overline{\mathcal{E}_{\text{scr}}}$, and

$$\text{SNR}_{\text{tmpl}} = 10 \log_{10} \frac{\mathcal{E}_{\text{true}}}{\overline{\mathcal{E}_{\text{scr}}}} \quad (1)$$

is bias-free, the same leakage entering numerator and denominator. Subtracting the floor from $\mathcal{E}_{\text{true}}$ before taking the per-sample root gives a bias-corrected phase-locked RMS in microvolts, which is the quantity that can be compared against the HEP amplitudes reported in the literature.

3 Results

Figure 1 shows the template of the showcase record with both intervals marked. Its bias-corrected phase-locked RMS is $10.9 \mu\text{V}$ over the QRS part and $2.62 \mu\text{V}$ over the HEP part, the latter 16.0 dB above the null ($p < 0.01$, 200 shifts). Across the 84 recordings with a resolvable template (of 436 analyzed; gate $\text{SNR}_{\text{tmpl}} \geq 10$ dB over the full window) the HEP-interval component is significant in 80, with median 10.0 dB in that interval alone, median $1.50 \mu\text{V}$ and maximum $7.71 \mu\text{V}$ RMS. It carries a median 28% of the removable phase-locked energy, and the spread is wide: the showcase record is at the favorable end, with only 8% of its removable energy past 200 ms, while at the other extreme a record’s phase-locked energy lies almost entirely there. The showcase should therefore be read as an illustration of the geometry, not as a typical cost.

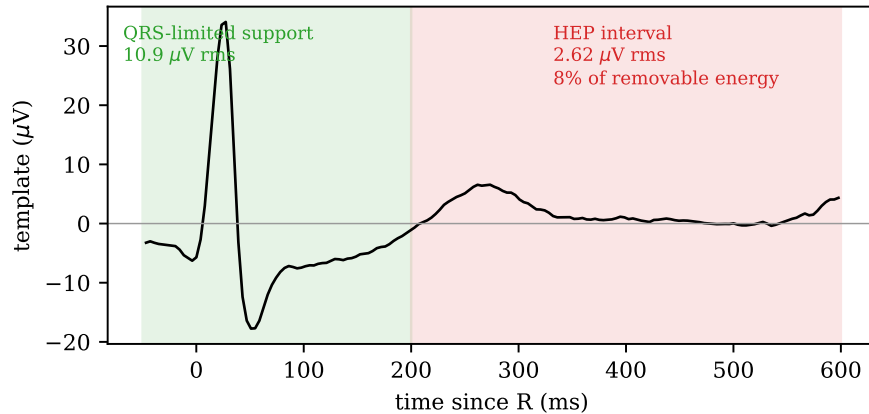


Figure 1: Non-causal template of the showcase record (ds005873, sub-070, ses-01, run-20, behind-the-ear channel, $N_r = 851$ learnable beats of a 600 s window) against time since the R-peak. Green: the QRS part $[-50, 200)$ ms retained by the QRS-limited template. Red: the HEP part $[200, 600)$ ms, annotated with its bias-corrected phase-locked RMS and its share of the removable phase-locked energy.

Both readings of these numbers matter, and neither on its own settles the question. Read as artifact, a phase-locked deflection of median $1.50 \mu\text{V}$, reaching $7.71 \mu\text{V}$, is worth removing, and is of the same order as reported HEP amplitudes—so leaving it in a continuous recording is not self-evidently safer than removing it. Read as neural signal, that same order of magnitude is exactly why the method must not be applied unexamined in an HEP study: whatever cortical response occupies the interval is inside the subtracted template and is not recoverable from it.

4 The QRS-limited template

Because the paper’s template is defined on the full cardiac cycle rather than on a fixed epoch, a restricted support costs nothing structurally: set $\hat{m}(j) = 0$ for every phase bin outside $[-50, 200)$ ms. The HEP interval is then returned to the output identical to the input.

The price is the phase-locked energy in that interval—a median 28% across the resolvable recordings, 8% on the showcase record, and on a minority of records nearly all of it. The main lobe used for the paper’s SNR_{tmpl} figures (50 ms pre to 150 ms post) lies entirely inside the retained support, so the QRS-limited template accounts for that figure in full.

5 Recommendation

- **QRS-limited by default** whenever the downstream analysis is not known in advance, or whenever heartbeat-locked cortical activity might be examined later. It removes the bulk of the artifact and forecloses nothing.
- **Full-window template** when heartbeat-correlated activity is unambiguously nuisance contamination—continuous wearable monitoring, decoding of non-cardiac rhythms—and the 200–600 ms interval will not be interpreted.
- **Neither** in a study whose dependent measure is the HEP. The estimator is not mis-specified there; it is being asked for something no timing-only method can provide.

Reproducing these numbers

```
python src/phase_hep_analysis.py --scan --max-runs 2 --duration-sec 600 \
    --out outputs/phase_hep_scan_full.json
python src/rerun_gated.py --out outputs/phase_hep_gated.json
python src/hep_figure.py --gated outputs/phase_hep_gated.json \
    --out paper/fig_real/fig_hep_template.pdf
```